



JASHORE UNIVERSITY OF SCIENCE AND TECHNOLOGY

Lab Report On

Setting Up Linux Environment

(i) Independent System

(ii) Dual Boot Mode

(iii) Using Virtual Box

Course: Operating System and System Programming Lab

Course Code: CSE 3102

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Title: Setting Up Linux Environment (Ubuntu 25.04 – Plucky Puffon)

Objective:

To install Ubuntu 25.04 (Plucky Puffon) Linux using the following methods:

1. Independent System Installation
2. Dual Boot with Windows
3. Virtual Installation using VirtualBox

Apparatus/Software Required:

- Ubuntu 25.04 ISO file
- 8 GB or larger USB drive
- Rufus (Windows) / Etcher (Mac/Linux)
- Oracle VM VirtualBox
- System with at least 4 GB RAM

Theory:

Linux can be installed in multiple modes. Ubuntu 25.04 is a modern, user-friendly distribution. Installing it provides hands-on experience with the Linux OS.

- **Independent System:** Linux is the only OS installed.
- **Dual Boot:** Both Linux and Windows are available.
- **VirtualBox:** Linux runs as a virtual machine within another OS.

Procedure:

(i) Independent System Installation:

1. **Download** Ubuntu 25.04 ISO from the official website.

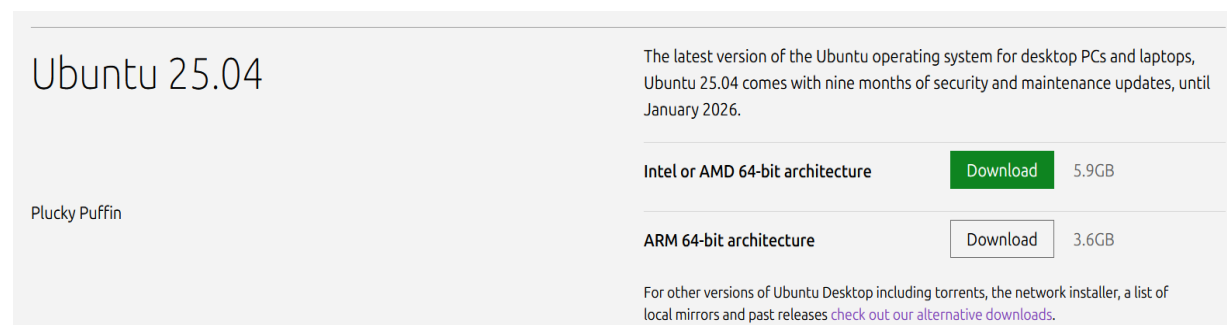


Fig:01

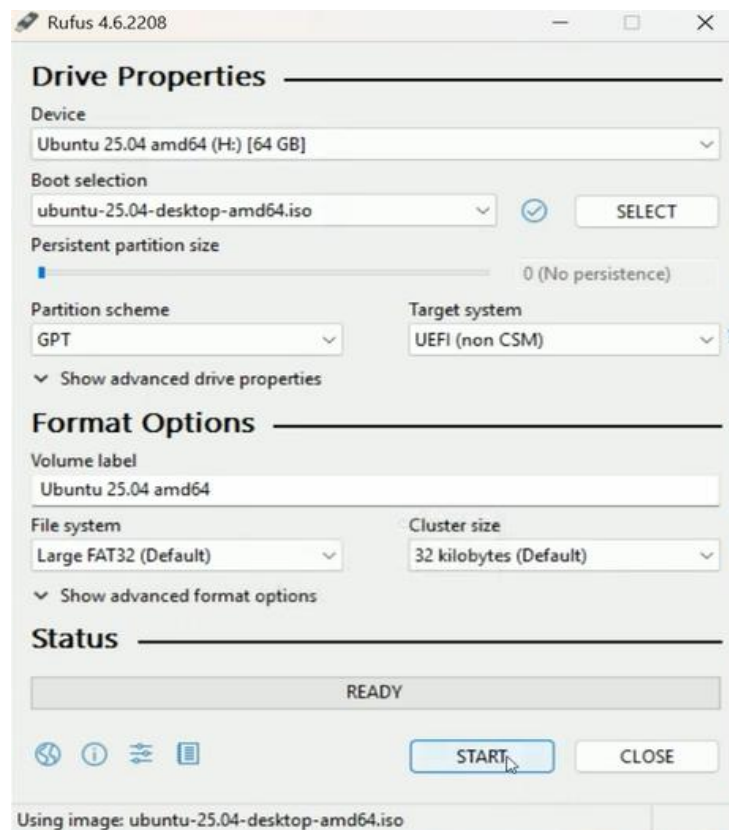
2. Create a Bootable USB (Using Rufus on Windows):

- Plug in an 8 GB or larger USB drive



Fig;01

- Open Rufus → select your USB drive
- Click "Select" and choose the downloaded Ubuntu ISO
- Set:
 - Partition scheme: GPT
 - File system: FAT32
 - Target system: UEFI (non-CSM)
- Click "Start", then "Write in ISO Image Mode"
- Wait for the process to finish



Fig;02

3. Restart Computer and enter BIOS (Repeatedly pressing ESC).

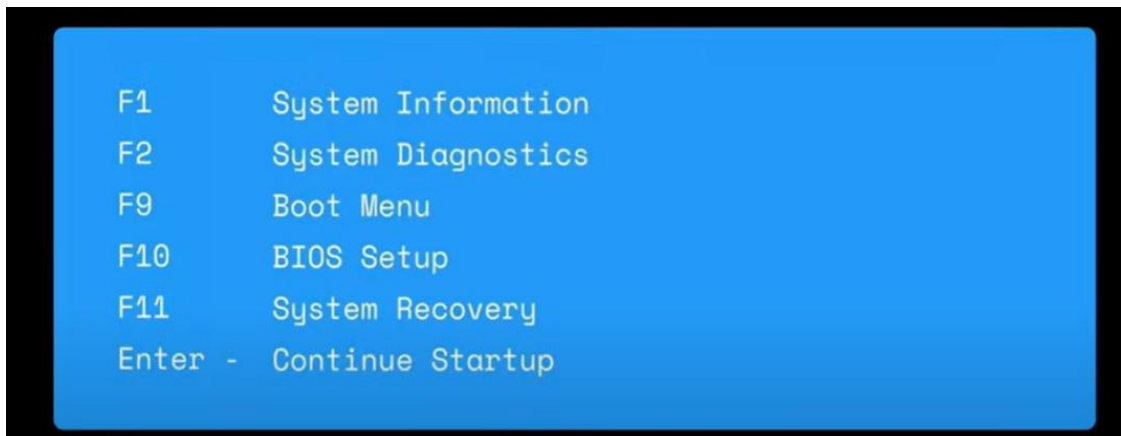


Fig:03

4. Enter Bios Setup(pressing F10) and **Disable** secure boot.



Fig:04

5. Boot into USB:

- Press the boot menu key (usually F10, F9, Esc, or Del) as the system starts
- Select the USB drive from the boot options

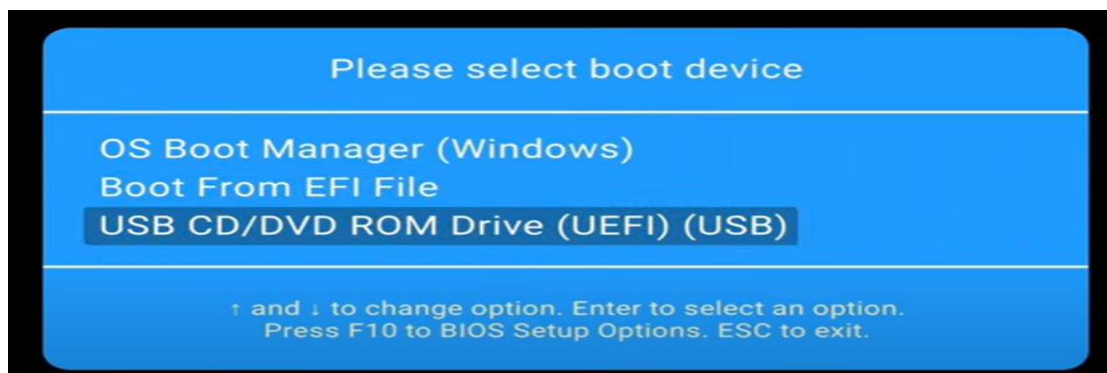


Fig:05

6. Install Ubuntu:

- Choose “Try or Install Ubuntu” → Click “Install Ubuntu”

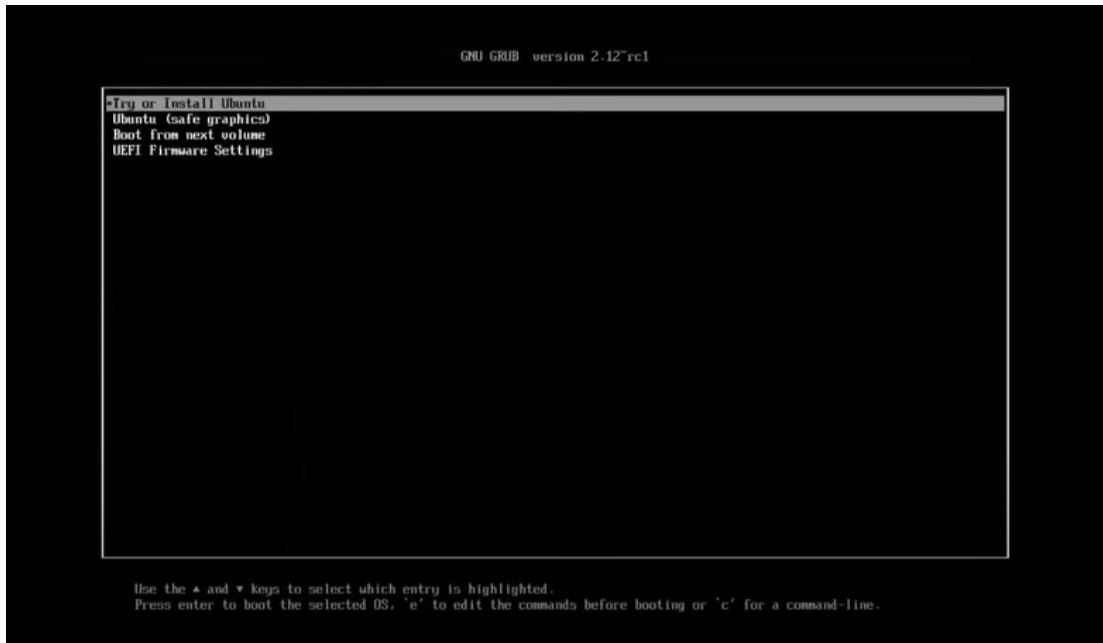


Fig:06

- Select language and keyboard layout
- Choose “**Normal Installation**” (recommended)
- Check both options for updates and third-party software
- **Installation type:** Select "Erase disk and install Ubuntu"
 - ⚠ This will delete all existing partitions and data

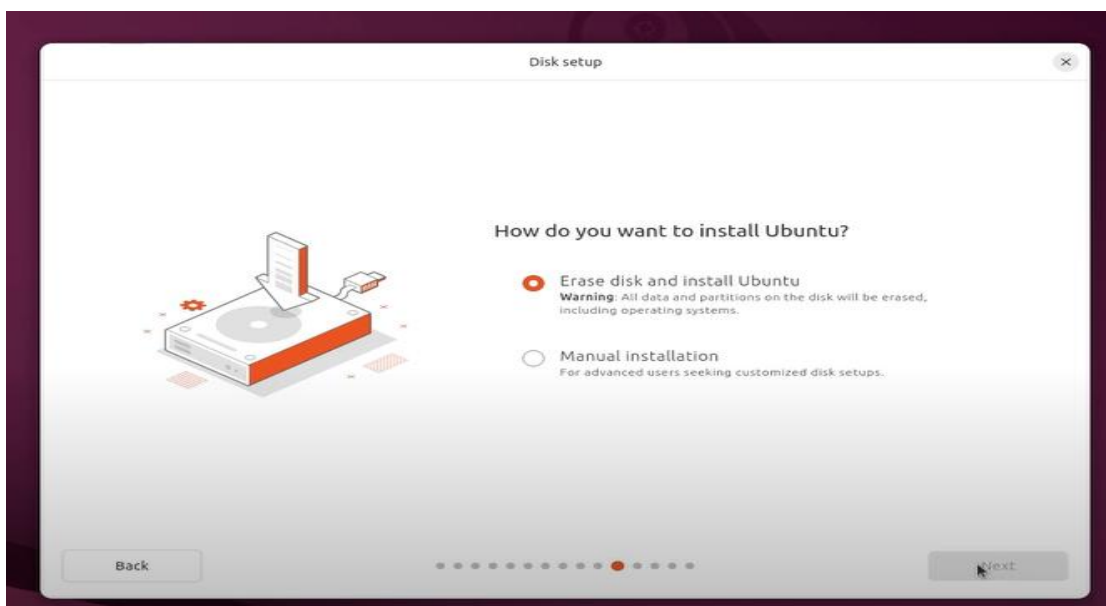


Fig:07

- Click **Continue**, review disk changes, and confirm

User Configuration:

- Create user account:
 - Your name, Computer name, Username, Password (and confirm)

Fig:08

- Click install

Fig:09

- Once installation is done, click **Restart Now**
- Remove the USB drive when prompted and System will boot into fresh Ubuntu 25.04

(ii) Dual Boot Mode Installation (Ubuntu + Windows)

1. Backup Your Data:

- Use external storage or cloud to back up files from Windows. In case something goes wrong during partitioning

2. Create Free Space in Windows:

- Open **Disk Management** (press Win + X → choose "Disk Management")
- Right-click on C: drive → choose **Shrink Volume**
- Shrink around **20 GB or more** for Ubuntu

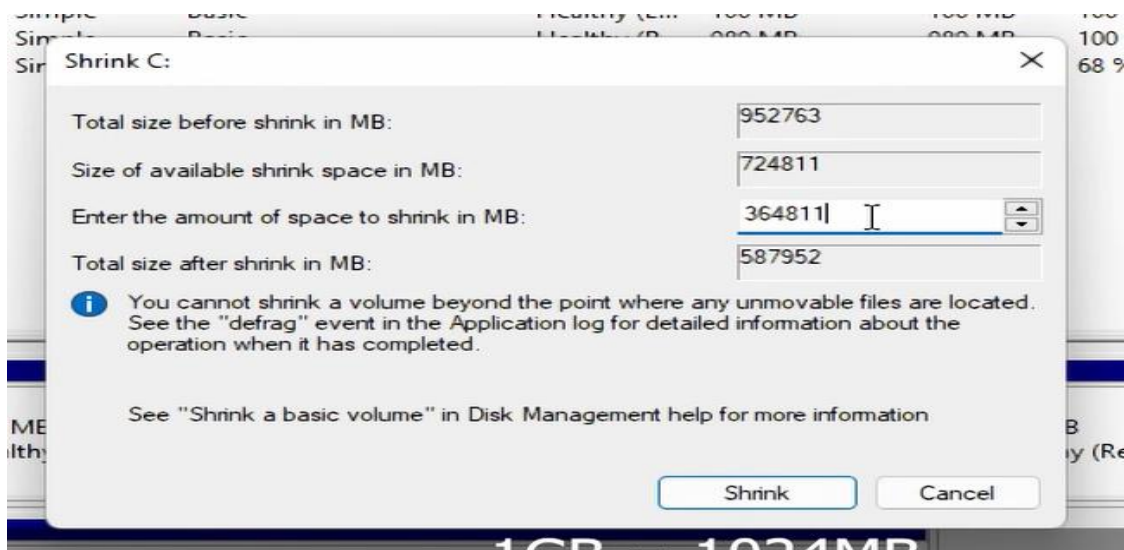


Fig:10

- Leave the space as **unallocated**

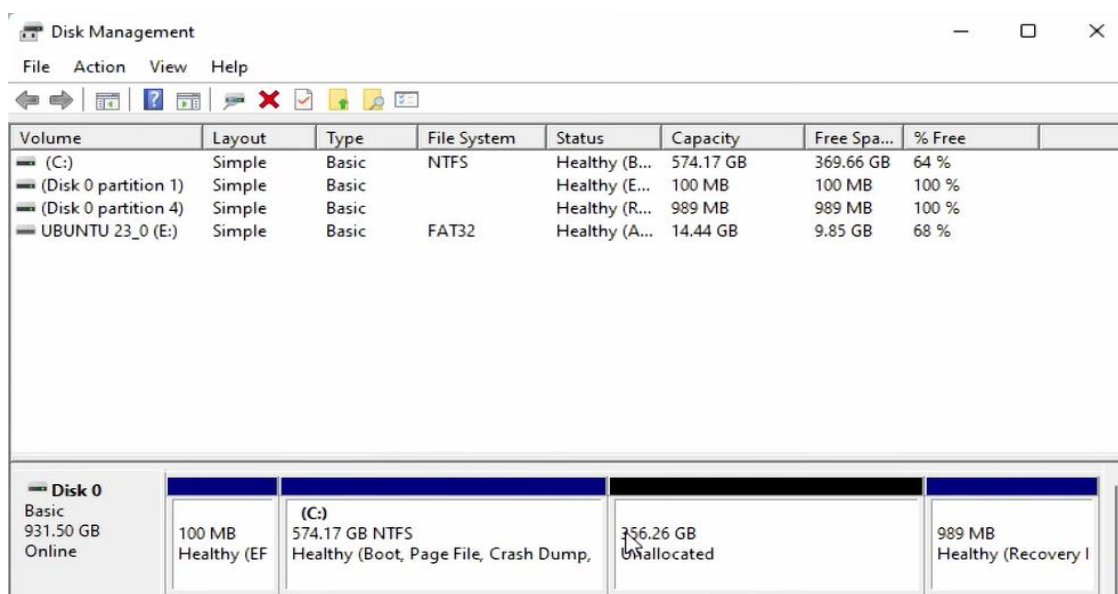


Fig:11

3. **Create Bootable USB (as above using Rufus)**
4. **Boot from USB: (Discribed before)**
 - Restart and enter boot menu (F12/F10/Esc)
 - Select USB drive to boot
5. **Install Ubuntu:**
 - Choose **“Install Ubuntu”**
 - Language, keyboard → select as before
 - Installation type: Choose **“Install Ubuntu alongside Windows Boot Manager”**

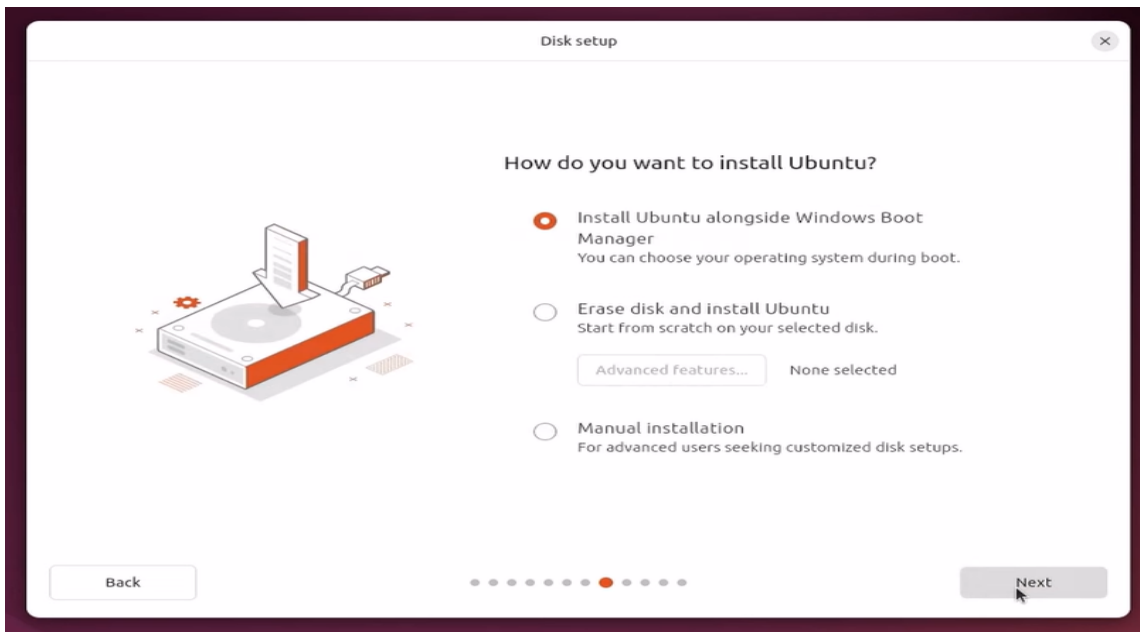


Fig:12

- Ubuntu installer will detect Windows and offer to install beside it
 - You'll see a slider to choose how much space for Ubuntu
 - Adjust partition sizes as needed
 - Click **Install Now**
6. **User Setup:**
 - Choose time zone, username, password
 7. **Finish Installation:**
 - After installation → **Restart Now**

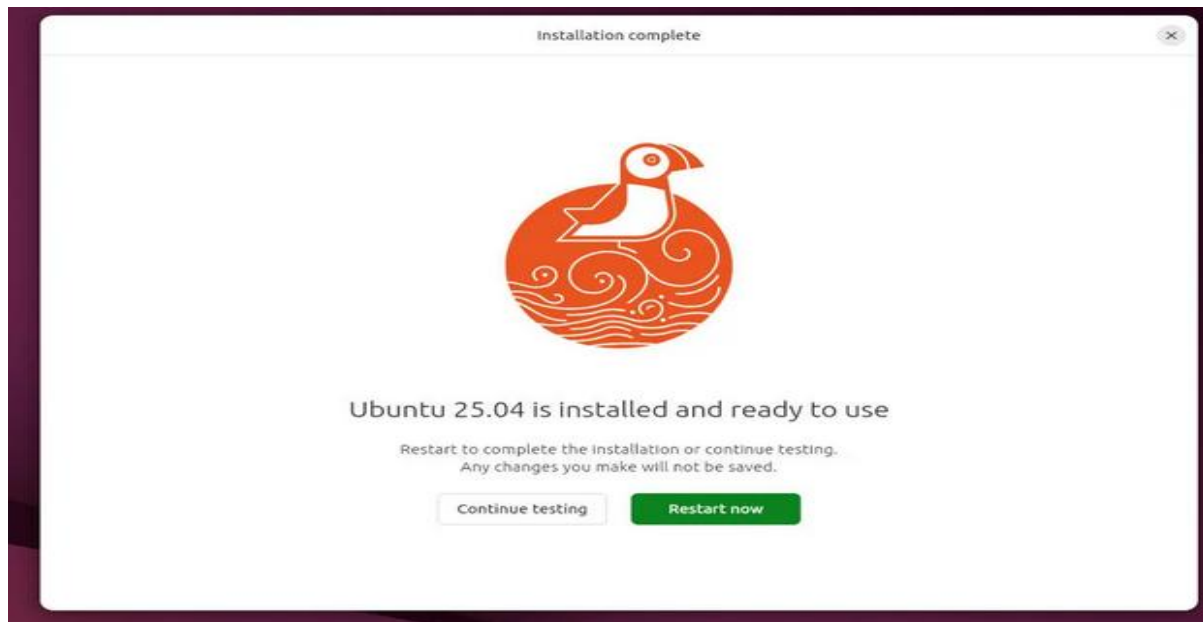


Fig:13

- On reboot, you'll see the **GRUB Boot Menu**
 - Options: Ubuntu / Windows Boot Manager
 - Use arrow keys to select OS and press Enter

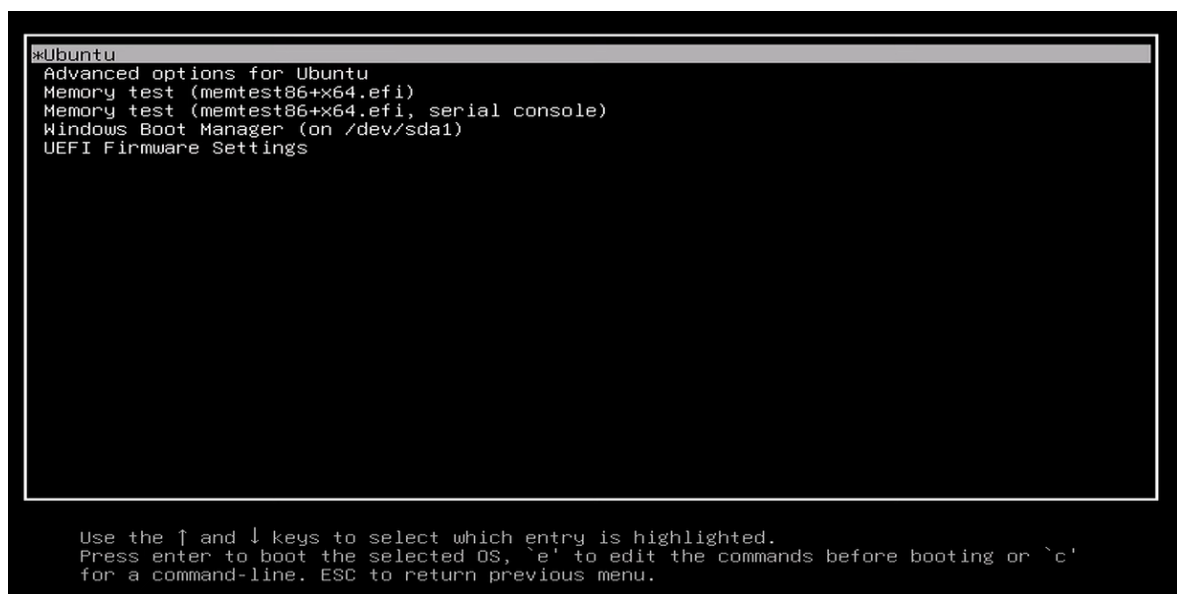


Fig:13

(iii) VirtualBox Installation:

1. Download & Install VirtualBox.



Fig:14

2. Open VirtualBox → click "New".

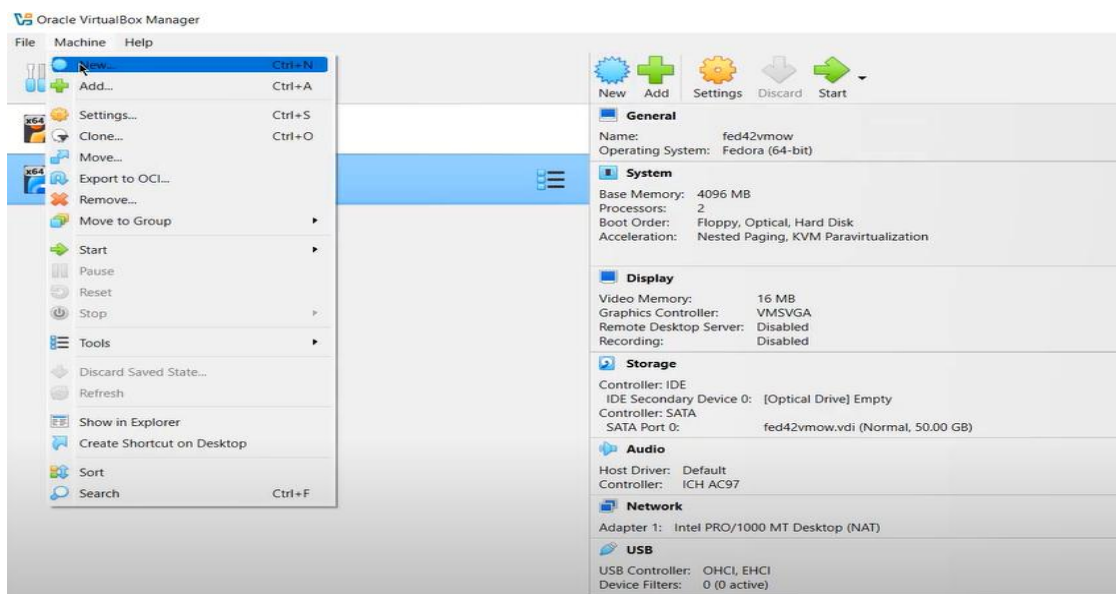


Fig:15

3. Set Name: Ubuntu 25.04, Type: Linux, Version: Ubuntu (64-bit).



Fig:16

- Assign at least **2 GB RAM** and **20 GB virtual disk**.

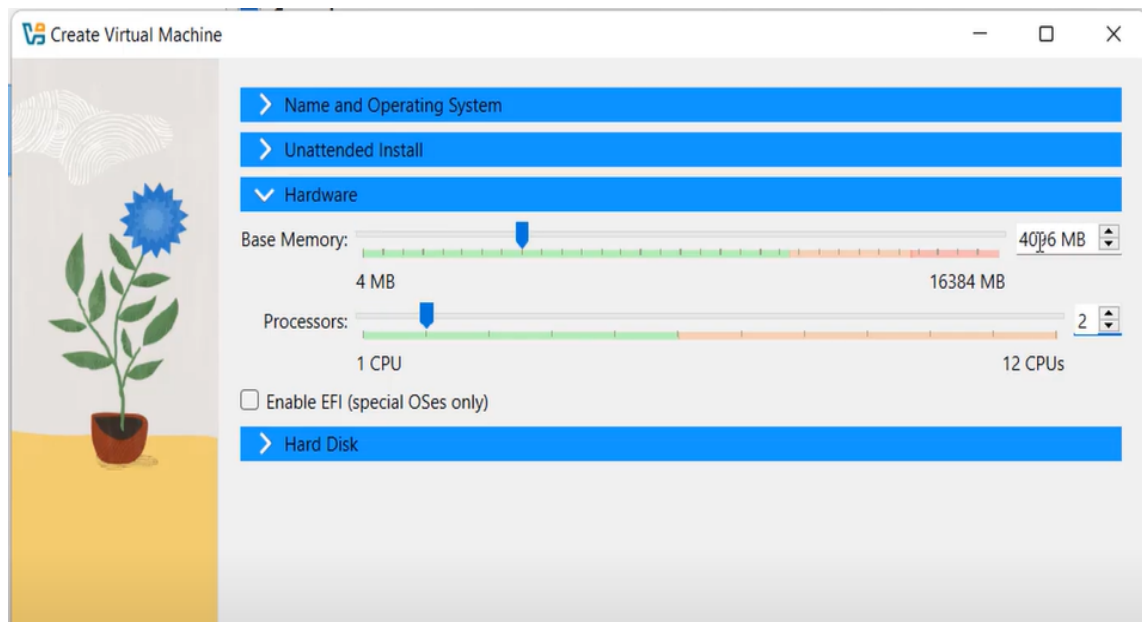


Figure:16

- Under settings → Storage → mount Ubuntu 25.04 ISO.

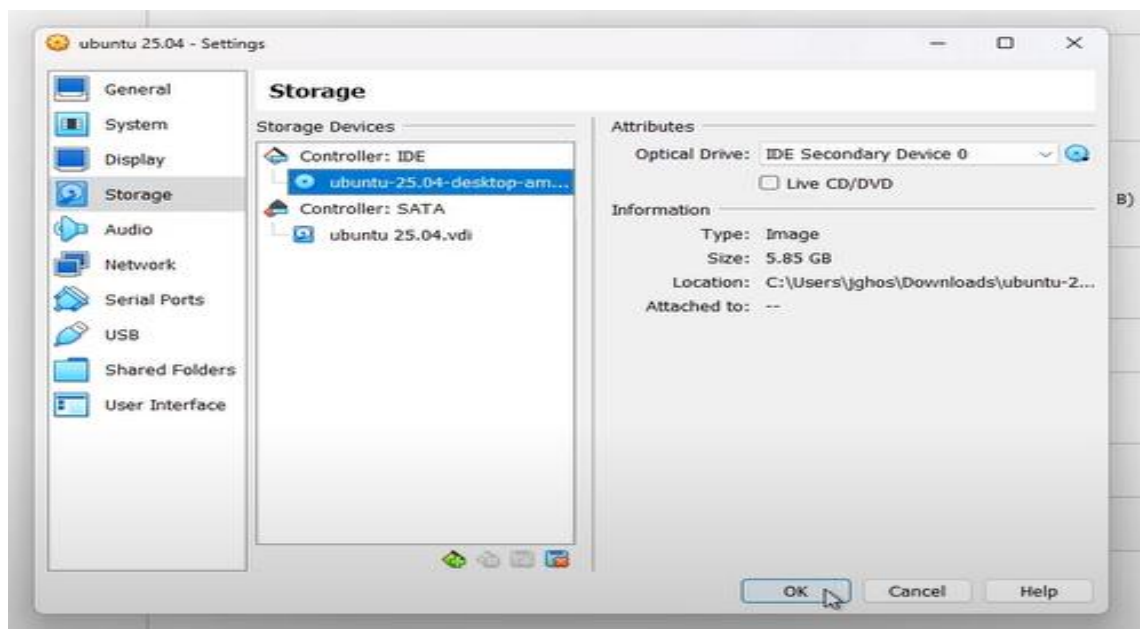


Figure:17

- Start the VM → follow Ubuntu installer steps.

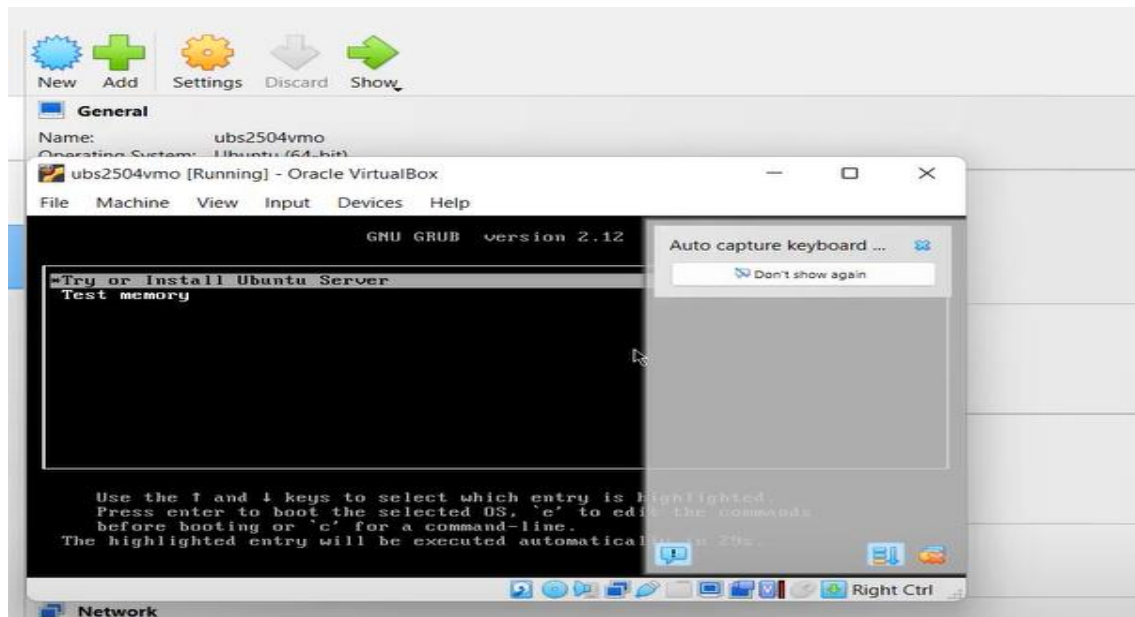


Fig:18

7. Complete Setup: Ubuntu is configured to run within VirtualBox, **following the steps outlined earlier**

Result:

Ubuntu 25.04 (Plucky Puffon) was successfully installed and tested in:

- Independent setup
 - Dual boot with Windows
 - VirtualBox environment
-

Conclusion:

All three methods give different advantages. Independent systems offer full performance, dual boot enables flexibility, and VirtualBox is safe for testing. Ubuntu 25.04 provides a modern, smooth Linux experience.